

Quantifying the Effects of Forest Management Strategies on the Production of Forest Values: Timber, Carbon, Oxygen, Water, and Soil

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Abstract Forest management practices alter forest structure quantified with ecosystem characteristics and values. In this paper, we utilized a forest management simulation model to assess the effects of three forest management strategies focusing on timber production, carbon sequestration, oxygen production, soil erosion, and water production of a forest management unit in Turkey. A forest simulation model “ETÇAPSimülasyon” was developed and used to project forest ecosystem development over 100 years under three forest management policies of timber-oriented forest management (TFM), multipurpose forest management (MFM), and no intervention (NI). The results showed that TFM strategy produced more timber and its net present value than MFM and NI strategies did. The amount of carbon sequestration and oxygen production potential was also found to be the highest with TFM strategy than with the MFM and NI strategies. Compared with the other strategies, however, NI strategy produced the highest amount of water production and soil losses over the planning horizon. The effects of a forest management strategy depend mainly on the initial forest structure, the rate of development and the level of forest management interventions. Therefore, forest dynamics under various management strategies should be explained before the final management decision. Understanding long-term effects

of any management strategies on forest structure will provide the basis for better reaching the management objectives.

Keywords Forest management · Forest ecosystem · Simulation · ETÇAP

1 Introduction

Forest ecosystems provide an assortment of values, such as timber, fuel wood, fodder, water and soil protection, carbon sequestration, oxygen production, recreation, aesthetics, biodiversity, and habitat for wildlife species [3, 15, 17]. A sustainable flow of these values are influenced by forest ecosystem characteristics, often defined in terms of plant species composition and their age class distribution, density, and development stages of various plant cohorts that are shaped by management interventions except for catastrophic events like fire, epidemic infestation [4, 17]. The purpose of forest management is to design interventions over time and space to accommodate and understand tradeoff among various forest values as desired by the society. Designing such interventions has been a daunting challenge, however, without the use of computer-based simulation and optimization models [3, 4, 16].

In recent years, forest management planning is extending its efforts to incorporate other forest ecosystem values such as carbon sequestration in addition to the traditional production goal. Sustainable forest management has thus broadened from sustainable timber management to ecosystem-based multipurpose forest management [4]. While many jurisdictions have embraced multipurpose forest management as the leading forest management paradigm, there is still room to explore forest simulation and optimization models aiming at multiple-use forest management objectives.

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Among studies about multipurpose forest management, some important forest ecosystem values such as amenity, recreation, soil protection, carbon sequestration, and biodiversity have been regularly incorporated into forest management plans [2, 3, 5, 6, 8, 9, 12–15, 19, 22, 23]. However, these studies have generally focused on the results based on broad level characterization of forest ecosystems such as age class structure or stand types (not individual stands) to predict the interactions among forest ecosystem values. A limited number of research initiatives using simulation models have presented the effects of ecological and environmental oriented forest management strategies on forest ecosystem functions such as carbon sequestration and biodiversity [18], recreational suitability, water purification, and air purification [17]. However, there have been very limited attempts to simulate the effects of various forest management policies on other forest goods and services such as soil erosion, timber, and water and oxygen production.

This study develops a forest simulation model “ETÇAPSimülasyon” to assist the formulation of forest management plans under various planning strategies including multiple forest values (Fig. 1). The primary objective of the research is

to examine the effects of three forest management strategies on forest ecosystem development. The second objective is to predict the long-term effects of these policies on some forest ecosystem values such as timber production, carbon sequestration, oxygen production, water production, and soil erosion.

2 Material and Methods

2.1 ETÇAPSimülasyon Model Description

ETÇAPSimülasyon is a stand-based forest-level decision support tool for assessing the effects of forest management interventions on forest dynamics and functions. The effects of the management regimes can be analyzed from the viewpoints of timber production, carbon sequestration, oxygen production, soil erosion, and water production. The model is developed and implemented with a collection of modules and components that could be reused when constructing new applications and models.

ETÇAPSimülasyon is a deterministic simulation model consisting of a number of primary components; data input, actions, and outputs. Data input includes all the procedures required to enter the spatial data about initial forest structure including yield tables, economic revenues and costs, silvicultural regimes, and management guidelines. The forest cover type map with its attribute data is also required to use stand specific information and to map out the spatial distribution of management decisions. The actions refer to the silvicultural prescriptions identified for each stand type and management policies for the whole forest. Prescription is a series of silvicultural treatments or management interventions to be applied to a stand type or a specific stand over a planning horizon. Model output provides forest performance indicators such as total timber production, water production, soil protected, carbon sequestered that show the temporal change for the forest ecosystem.

Defining management strategies is necessary before forecasting forest development over time. A management strategy consists of a set of rules for queuing stands for harvest and a set of levels at which these activities will be carried out [11]. A management strategy also accommodates the option of planning horizon length, planning period length and harvesting method (volume or area control). The rules are used to control the numerical and geographic distribution of management actions. Harvest rules can be combined to control the harvest by queuing stands for harvest based on both stand condition and geographic arrangement. Harvesting rules available in ETÇAPSimülasyon are harvesting stands with the highest primary volume loss (i.e., mortality loss), harvesting stands with the oldest ones, harvesting stands with the highest increment loss, and harvesting stands with the highest volume per hectare.

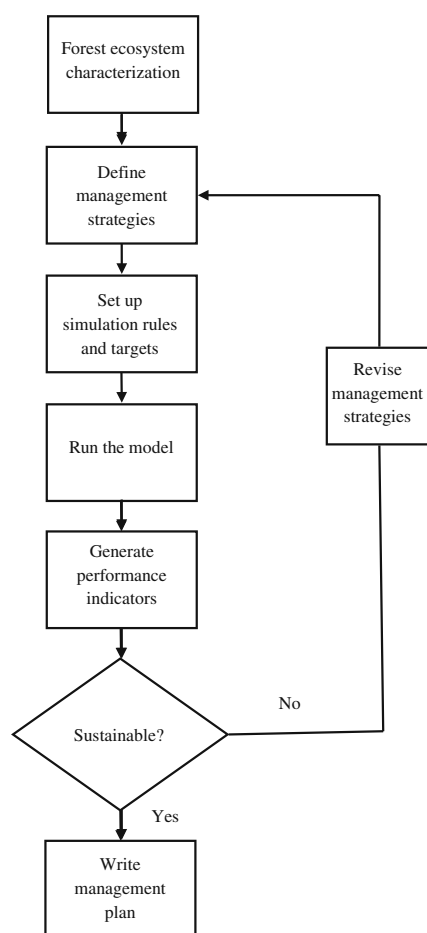


Fig. 1 A typical process of forest management simulation model

ETÇAPSimülasyon forecasts forest development over time in response to a management strategy. Forest response to management actions, as controlled by queuing rules, is forecast on an iteration-by-iteration basis. When harvesting is completed in a period of one iteration, the forest is simulated by aging all the stands one period ahead. Harvested stands or future stands are assigned to the first age class and follow the development pattern of empirical yield tables. Unharvested stands are assumed to follow their predecessor yield curve after breakup and reassigned to the first age class. In case there is no yield curve, the stand growth is projected based on the relationship (i.e., ratio) between the current stand parameters and future stand parameters gathered from the empirical yield table. The harvesting process is repeated until the requested time horizon is reached or the available periodic wood supply is exhausted. Projections are conducted for a planning horizon (i.e., 100 years), divided into periods (5, 10, 15, etc.). In case a given harvest level cannot be sustained, the model reduces the harvest level by a specified percentage and runs it again until a threshold level is reached.

The basic spatial unit represented in ETÇAPSimülasyon is a user-defined *analysis area*. Each analysis area consists of a number of forest stands, the smallest management unit, defined based on forest conditions such as species compositions, development patterns, crown closure and site productivity. A stand by itself would also be identified as an analysis area. The model provides means for alternative silvicultural regimes for each analysis area defined by the users. One or more silvicultural regimes can be prescribed to an analysis area. Silvicultural regimes for each analysis area are designed based on inventory data, forest-level and stand-level-defined forest treatment options.

The functionality provided by the model may be interpreted as the generation and real-time visual display of different management strategies or alternatives carried out over time. The results can be displayed in the formats of tables, graphics and forest thematic maps. In order to increase the visual representation of the forest landscape, ETÇAPSimülasyon provides the capability for viewing graphics, tables, and maps associated with actual or simulated forest areas in relation to potential management regimes.

2.2 Growth and Yield Projection

A stand simulation model within the *ETÇAPSimülasyon* model was developed to project the growth of each existing stand as there was no growth and yield model constructed for the species of the forest. So the growth and yield of existing stands over the following period are predicted based on the relationship between the inventory data and the empirical yield table. The stand simulation model

assumes that the growth of a particular stand will follow the similar definable and predictable development pattern of the empirical yield table. Here, the current growth of each stand was measured with timber cruising or inventory data. Specifically, the current growth rate of each stand is related to the growth rate of the same stand in the yield table to predict the future growth of each stand. Specifically, the growth of current stands is predicted based on the relations between the stand parameters gathered from the field survey, and the parameters of stands in empirical yield tables. For example, the basal area of a stand in the next period is predicted based on the ratio of current basal area to the optimal basal area of the same stand type. Empirical yield tables are used to predict the growth of the regenerated stands in a forest. ETÇAPSimülasyon model utilizes user-supplied yield tables. Thus, the model accommodates all stand-level information related to each stand, such as basal area, growing stock, increment, and number of tree associated parameters of a stand. The model is not, however, limited to the number of yield tables that can be included.

2.3 Financial Analysis

The financial information associated with various timber products or assortments is used to calculate all the costs and revenues derived from the harvest, thinning, regeneration, and reforestation activities. Timber harvest financial data include the stumpage price for timber, harvesting costs and wood assortments revenues. The model assumes that values and costs can change throughout the simulation period (i.e., planning horizon), and uses the Net Present Value economic approach. This allows for the calculation of actual net revenues for each year of the simulation. A certain discount rate, based on the economic condition of forestry sector in the country, is used to compute the revenues and costs projected to today's economic situation. The users need to provide the discount rate appropriate for the planning situation under the given jurisdiction.

2.4 Carbon Sequestration

In *ETÇAPSimülasyon* model, carbon sequestration in successive periods was determined by accumulating the amount of timber biomass of the forests and subtracting the biomass removed from the ecosystem by various reasons such as harvesting. The following equation measuring the net carbon sequestration in the t th period was used in this study [6].

$$CS_t = [\gamma(V^t - V^{t-1} + H_t) - CE_t] \quad (1)$$

where γ is the proportion of carbon contained in timber biomass, CS_t is the amount of carbon sequestration in

period t , CE_t is the amount of carbon emission in period t , H_t is the volume harvested in period t and V^t is the volume of forest inventory at the end of t th period.

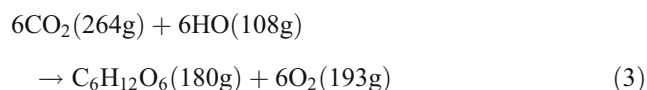
Forest timber biomass and carbon storage for each stand was estimated using species-specific biomass expansion or conversion factors from the literature [3, 4, 14]. The carbon emissions from various forest timber assortments were also taken into consideration and estimated in this study based on the lifetime of each wood product for each stand [3, 4, 14]. Decomposition rates were estimated by applying the methodology proposed by [20].

$$Cp_{mt+1} = Cp_{mt} \alpha (1 - a_m) \quad (2)$$

Where, Cp_m is the carbon stored in a wood product m at time t , α_m is the proportion of the product that decomposes each year.

2.5 Oxygen Production

Forests produce oxygen in response to the consumption of CO_2 as part of photosynthesis. In ETÇAPSimülasyon model, we adopted a method similar to [1, 3, 7] based on the formula of photosynthesis and respiration. The formula of photosynthesis and respiration is as follows:



According to Eq. 3, forest absorbs 264 g CO_2 for the production of 162 g dry material. In other words, it needs 1.63 g CO_2 and releases 1.2 g oxygen to form 1 g dry material. Carbon is converted to CO_2 equivalent by multiplying by 44/12 (3.6667). Oxygen balance is calculated like carbon balance because oxygen is consumed when cutting residues, deadwood and forest products decomposed.

2.6 Soil Erosion

The service or the value of soil erosion for forest ecosystems varies significantly with the differences in some characteristics of forest vegetation such as tree species, basal area, mean diameter of stand, standing timber volume, and the number of stems. In the current version of ETÇAPSimülasyon model, the amount of soil protected by forest ecosystems was estimated according to different soil erosion models. All these models estimate the amount of soil loss in relation to stand basal area. Based on the modularity of ETÇAPSimülasyon model, however, a new soil erosion model can be incorporated into the model easily. In this study, we selected the soil erosion model developed for conifer forests including *Pinus sylvestris* stands in the northeastern forest ecosystems [24]. In

determining soil losses of forest stands, [24] used the Universal Soil Loss Equation which includes some important parameters such as rainfall erosivity factor, soil erodibility factor, slope length factor, slope factor and cover management factor. The equation is as follows;

$$SL = 30.437 * e^{(-0.0488 * BA)} (R^2 : 0.55, SE : 0.69) \quad (4)$$

Where, SL is the amount of soil loss (ton/ha/year), BA is the residual stand basal area (m^2/ha), and e is the 2.71828.

2.7 Water Production

The water quality and quantity of forest ecosystems have been affected by a number of stand parameters such as tree species, crown closure of stands, basal area, mean diameter of stand, number of stems, standing timber volume, and leaf area index of trees. In the current version of ETÇAPSimülasyon model, the amount of water production of forest ecosystems was estimated according to a few water production functions. Again, based on the modularity of ETÇAPSimülasyon model, however, a new water production models can easily be incorporated into the model. The water production response function used in this paper is equation developed for *P. sylvestris* stands in the case study area by [21]. The model equation for the water that runs off is as follows;

$$WP = 1797.97 * e^{-0.0196 * BA} (R^2 : 0.50, SE : 0.19) \quad (5)$$

Where, WP is the annual water production (ton/ha), BA is the residual stand basal area (m^2/ha), and e is the 2.71828.

3 Case Study Area

The study area of Yalnızçam Forest Planning Unit is situated in Ardahan city in the northeastern corner of Turkey. The area consists of primarily high mountain forests and scattered settlements such as villages and upland shelter lands. The altitude varies between 1,800 and 2920 m with an average slope of 23%. Naturally, the region is covered by *P. sylvestris* L., the most widely distributed species in the world and the species is in its most southern site of the world distribution. The study area covers an area of 44,680 ha of which 6,753 ha is forestland. The total number of stands in forested areas is 944 with an average size of 2.5 ha. Forest stand types in the study area are grouped according to the mixture of tree species, development stage, and crown closure.

3.1 Forest Management Strategies

Three major forest management strategies were developed to examine various forest management planning approaches.

These include timber-oriented forest management (TFM), multipurpose forest management (MFM), and no intervention (NI).

Timber-Oriented Forest Management This strategy attempts to maximize total amount of timber production. All forest stands are subject to harvesting. Minimum rotation ages or cutting ages for Scots pine stands were defined as 100 years in good sites and 120 years in poor sites. Maximum rotation age, however, was set to 180 years. These age limits define the operability window of the stands and are reference cases as adopted from the Turkish Forest Management Guidelines in timber-oriented forest management planning approach. The management guidelines define the minimum operable ages for each tree based on site quality on each ecological region in Turkey. Commercial thinning action was designed for the stands whose ages range from 30 years to 100 years.

Multipurpose Forest Management The primary objective of this strategy is to maximize total amount of timber production. While it seems similar to the former strategy, here not all stands are subject to harvesting though. Forest ecosystems were stratified into seven management units (forest use areas) according to the various forest values (functions): timber production, ecosystem rehabilitation, high mountain forest ecosystem, biodiversity conservation and monitoring, water and soil protection forest, aesthetics-recreation, and social conflict forest areas. Minimum rotation ages in management units of timber production, ecosystem rehabilitation, high mountain forest ecosystem, and aesthetics-recreation were defined as 100 years in good sites and 120 years in poor sites, and maximum rotation ages were 180 years. The stands can be thinned in 30, 50 and 70 years. Minimum and maximum rotation ages in the other management units were defined as 180 and 250 years, respectively and the stands within the management units can be thinned in 40, 70 and 100 years.

No Intervention It also means “Do Nothing”. All forest stands are subject to growing without harvesting or thinning activities.

Forest management strategies were implemented on a periodical basis and forest outputs were controlled to be sustained over the planning horizon. Such approach prevents, of course, all available stands to be harvested or treated in a specific period, allowing forest structure to be regulated over time. However, as there is no tradeoffs among the periods the model cannot guarantee the optimal solution. Another model called ETÇAPOptimization has been developed by the same authors to overcome such shortcomings.

In preparing forest management strategies for three approaches, some management activities and assumptions are provided as followings:

- The planning horizon is 100 years and the planning period is 10 years.
- All forest values and stand characteristics are estimated at stand level.
- All forest stands may be cut exactly once over a planning horizon when they reach the minimum cutting age in the TFM and MFM strategies.
- Regeneration is assumed to follow immediately after harvesting.
- Regenerated areas are assumed to develop according to empirical yield tables.
- Growth and yield projection of actual stands is forecast according to typical simulation of growth potentials of stands due to the unavailability of growth and yield models.
- All stand parameters and forest values are estimated at the midpoint of each period.
- Even flow of timber production and management policy in association with ending inventory constraint are applied to the TFM and MFM strategies.

On the other hand, the model has the potential to enable the forest managers to satisfy various objectives and outputs. Such advantages of the model, developing alternative forest management strategies from the model and their applicability are left to the preferences and priorities of decision makers. Here, the modeling exercise focuses on the understanding of forest dynamics through comparisons of model results from different forest management strategies with various objectives and constraints.

4 Results and Discussions

TFM strategy produced the highest amount of timber volume and also provided more net present value (NPV) of timber than MFM and NI strategies did (Figs. 2 and 3). These figures also show the periodical timber productions and the NPVs of timber for each forest management strategy over the planning horizon, respectively. In the TFM strategy, the total cumulative amount of timber production and the NPV of timber were three million m³ and \$35 million, respectively, at the end of the planning horizon. In strategy, MFM compared with TFM strategy, total cumulative timber production and the NPV of timber decreased by about 30% and 28.8%, respectively.

With regard to the water production, NI strategy produced more amount of water than TFM and MFM strategies did (Fig. 4) as the latter two strategies regenerated

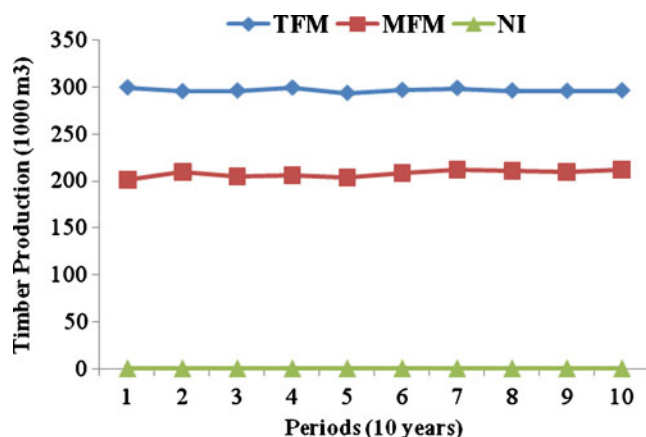


Fig. 2 Timber production per 10-year period of forest management strategies

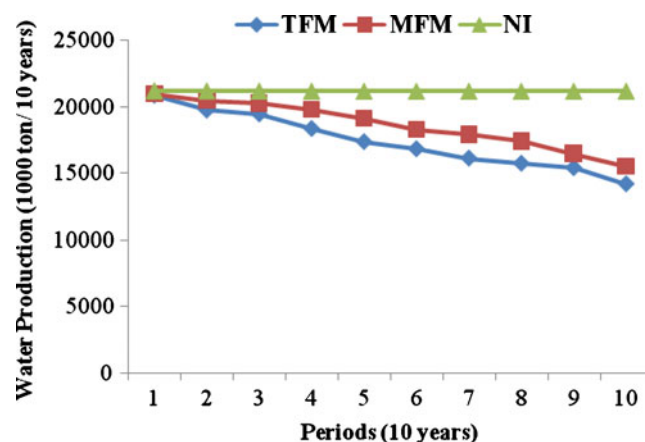


Fig. 4 Water production per 10-year period of forest management strategies

or planted open areas over time. Total water productivity from the TFM, MFM, and NI forest management strategies at the end of the planning horizon was 174, 186, and 212 million tons, respectively. Among the forest management strategies, however, NI strategy produced the highest soil losses (Fig. 5). Around 2.4 million tons of total cumulative soil losses occurred in the TFM strategy. Compared with TFM strategy, soil losses in the strategies of MFM and NI decreased by about 12.7% and 40%, respectively. In all strategies, water production and soil loss decreased over time when the forest is simulated under TFM and MFM strategies. The water production and soil loss stayed, however, relatively the same under NI strategy. As expected, increasing basal area caused more soil protection and less water production (Figs. 4, 5, and 6).

The periodical flow (tons/10 years) of oxygen production and carbon sequestration in all strategies were illustrated in Figs. 7 and 8, respectively. At the end of the 100 years of planning horizon, TFM strategy sequestered more carbon (976,300 ton as the total cumulative) than

MFM strategy (561,800 ton as the total cumulative) and NI strategy (158,000 ton as the total cumulative) did. Similarly, the total amount of cumulative oxygen production at the end of the planning horizon was the highest in the TFM strategy (2.6 million tons). The reductions in oxygen production with MFM and NI strategies were about 42% and 84%, respectively. While carbon sequestration and oxygen productions with the NI strategy decreased over time, these values for the TFM and MFM strategies first decreased with during the first 30 years of the simulation, and then increased substantially during the next 50 years. This result could be explained with the dynamics of growing stock (standing timber volume) as growing stock of forest ecosystems is an important forest performance indicator in analyzing the effects of forest management practices on forest structure and its functions. The growing stock of the TFM strategy was greater than those of the MFM and NI strategies at the end of the planning horizon (Fig. 9). As mentioned previously, carbon sequestration and oxygen production of forest ecosystems are related to

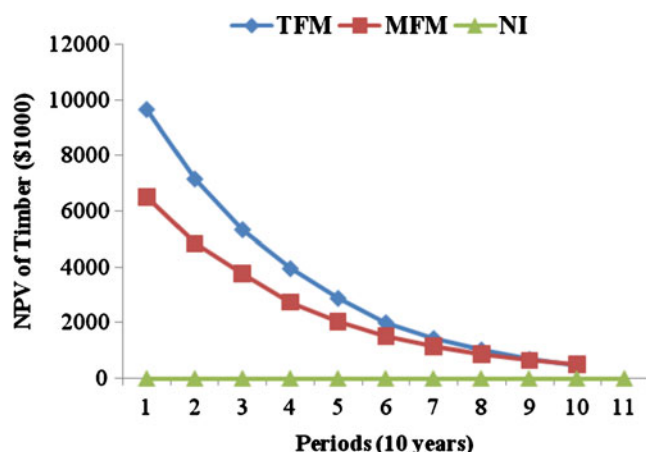


Fig. 3 NPV of timber per 10-year period of forest management strategies

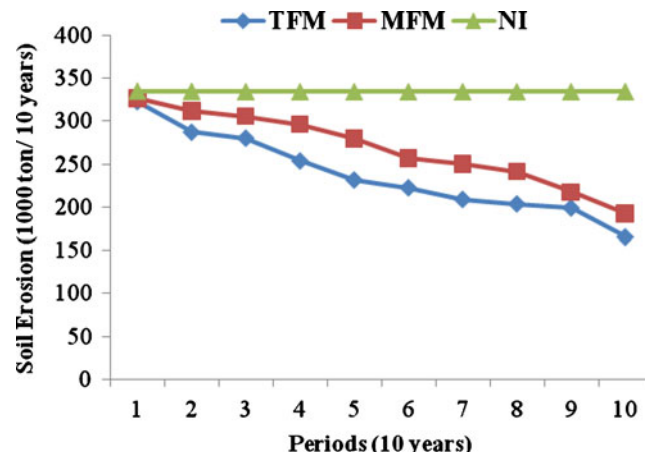


Fig. 5 The amount of soil losses over time for three forest management strategies

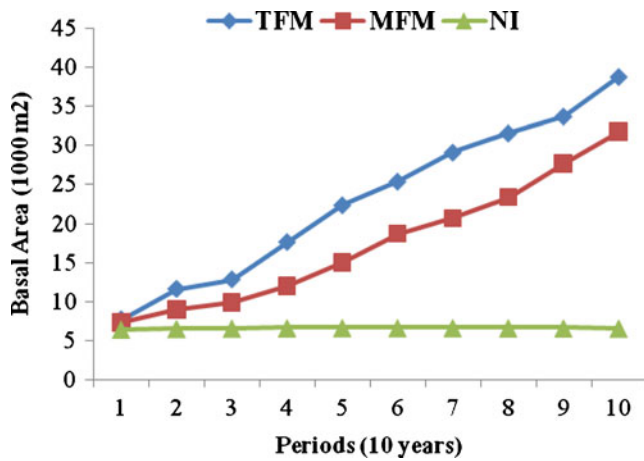


Fig. 6 The change in basal area under three management strategies

timber and biomass growth rates of forest ecosystems. Plant growth rates determine how rapidly carbon is removed from the atmosphere [4, 10]. According to model solutions, timber growth of forests in the TFM and MFM strategies was quite small in the first periods. When the trees are removed for timber production, and then replaced with seedling stock there is a lag phase as the trees develop and build their canopies.

5 Summary and Conclusion

This study developed and used a stand-based forest-level simulation model of *ETÇAPSimülasyon*. The model simulates the development of a forest ecosystem and the long-term effects of various management activities on the forest ecosystem structure and its functions over time. Simulation has been one of the most widely used quantitative techniques in forest management to understand forest dynamics. Simulation models, however, generate management plans and provide solutions without a guarantee of optimality. In this study, a number of forest ecosystem values (timber

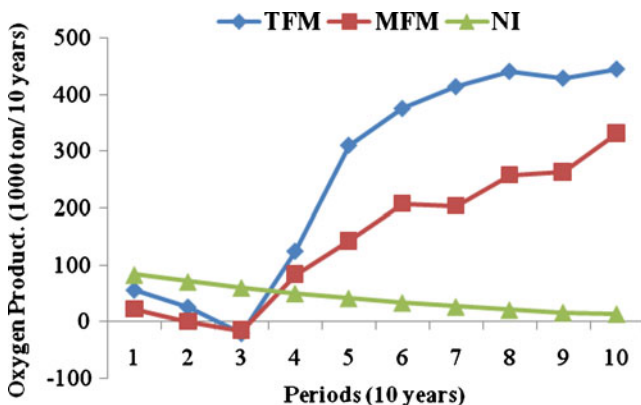


Fig. 7 Net oxygen production per 10-year period of forest management strategies

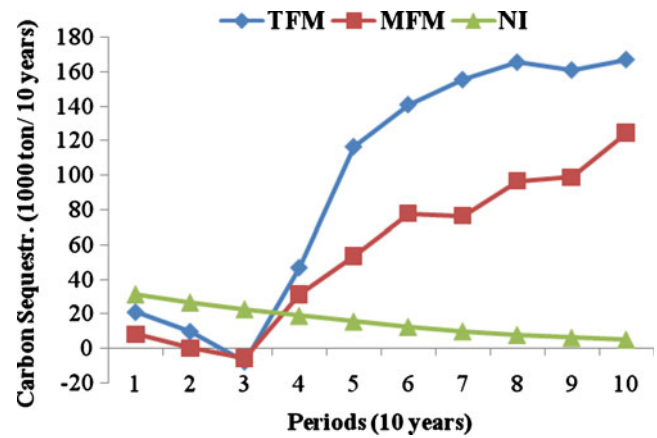


Fig. 8 Net carbon sequestration per 10-year period of forest management strategies

production, soil erosion, water production, carbon sequestration, and oxygen production) were successfully incorporated into the forest management planning process with timber-oriented forest management, multipurpose forest management and no intervention management strategies in mind.

The specific scope and characteristics for the *ETÇAPSimülasyon* model in this research can be summarized as followings:

- It is a general purpose long-term management model that simulates the dynamic behavior of forest ecosystems with ecological and financial understandings of forest management activities.
- The smallest and basic management area for this model is a stand. There is no limit in the number of stands to be included in the simulation. However, stands can also be grouped into homogeneous units (called analysis areas) based on a user-defined criteria.
- *ETÇAPSimülasyon* is a deterministic model for even-aged stands, as it stands. However, new version is about to be developed to accommodate un-even-aged stands.

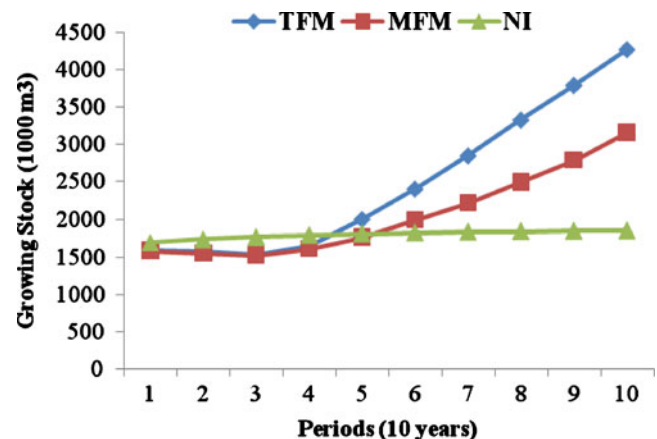


Fig. 9 Growing stock per 10-year period of forest management strategies

- The dynamics of growth is driven by yield tables and forest inventory data provided by the user.
- The model provides area and volume control management policies and the user predefines the target volume or area to be harvested or treated.
- The focus of the forest management component of the model is to sustain timber supply over time. It has the capability to manage different harvesting priorities. Harvesting is only allowed in stands within the operability window of each species.
- There is no limit in the number of forest management strategies and silvicultural regimes to be prescribed.
- The model simulates long-term effects of management prescriptions on some forest ecosystem values provided by the user.
- The user can create, edit, save, and retrieve the different input files needed to accomplish a simulation run. The user is able to design customized reports. The results can be obtained in the formats of tables, text, figures and maps.

The current structure of the forest ecosystem that results from historical management regulations may have certain impacts on various timber and non-timber forest goods and services. The quality and quantity of all forest goods and services are influenced by forest characteristics such as stand structure, spatial distribution, tree species composition, and developmental stages as well as management policies. Various forest management activities have affected the forest structure and its functions. Therefore, if a forest management plan is to meet public demands for certain forest values; the long-term effects of alternative forest management strategies including various policy constraints and targets should be identified and understood well.

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